

DESIGN DOCUMENT

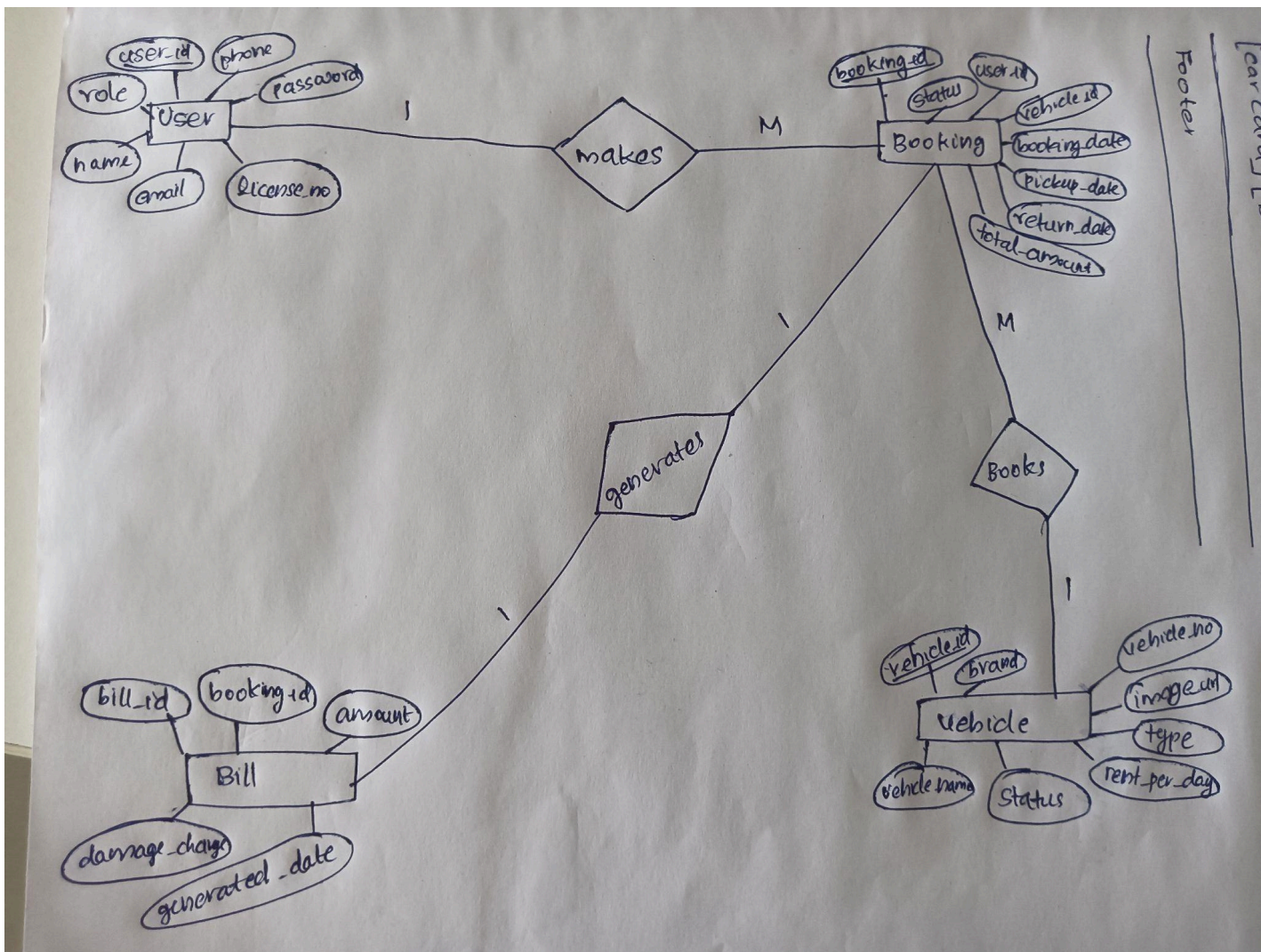
RentRide: Online Vehicle Rental and Booking Management System

1. Project Overview

1.1 Introduction

RentRide is a web-based vehicle rental management system that enables customers to browse available vehicles, book vehicles online, and manage their bookings. The system also allows administrators to manage vehicle inventory, customer information, vehicle returns, and revenue tracking.

The application is designed to simplify vehicle rental operations by automating booking validation, availability checking, and billing processes.



2. System Architecture

The system follows a three-tier architecture.

Presentation Layer (React)



Business Layer (Spring Boot)



Data Layer (MySQL)

Presentation Layer

Responsible for:

- User Interface
- Form Validation
- API Communication

Technology:

- React.js
- HTML
- CSS
- JavaScript

Business Layer

Responsible for:

- Authentication
- Business Logic
- Booking Processing
- Billing Calculation

Technology:

- Java
- Spring Boot
- Spring Security
- Spring Data JPA

Data Layer

Responsible for:

- Data Storage
- Data Retrieval
- Relationship Management

Technology:

- MySQL
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3. Technology Stack

Component	Technology
Frontend	React.js
Backend	Spring Boot
Database	MySQL
Authentication	JWT
ORM	Spring Data JPA
Build Tool	Maven
API Testing	Postman
Version Control	Git & GitHub

4. Module Design

4.1 Authentication Module

Features

- User Registration
- User Login
- JWT Token Generation
- Role-Based Authorization

Roles

- ADMIN
 - CUSTOMER
-

4.2 Customer Module

Features

- View Profile
 - Update Profile
 - View Booking History
 - View Rental Bills
-

4.3 Vehicle Module

Features

- Add Vehicle
- Update Vehicle
- Delete Vehicle
- View Vehicle Details
- Search Vehicles

Vehicle Attributes

- Vehicle Name
 - Vehicle Number
 - Brand
 - Type
 - Rent Per Day
 - Status
-

4.4 Booking Module

Features

- Create Booking
- Cancel Booking
- Check Vehicle Availability
- View Booking History

Booking Workflow

Select Vehicle



Select Dates



Availability Check



Calculate Amount



Booking Confirmed

4.5 Billing Module

Features

- Automatic Bill Generation
- Rental Cost Calculation
- Damage Charge Calculation

Formula

Total Amount =
(Rent Per Day × Rental Days)

- Damage Charges

4.6 Dashboard Module

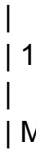
Features

- Total Vehicles
- Available Vehicles
- Active Bookings
- Completed Bookings
- Total Revenue

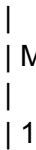
5. Database Design

Entity Relationship Overview

User

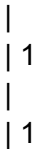


Booking



Vehicle

Booking



Bill

Users Table

Field	Type
id	Long

name	String
email	String
password	String
phone	String
licenseNumber	String
role	String

Vehicles Table

Field	Type
id	Long
vehicleName	String
vehicleNumber	String
brand	String
type	String
rentPerDay	Double
status	String
imageUrl	String

Bookings Table

Field	Type
id	Long
userId	Long
vehicleId	Long
bookingDate	Date
pickupDate	Date
returnDate	Date
totalAmount	Double
status	String

Bills Table

Field	Type
id	Long
bookingId	Long
amount	Double
damageCharges	Double
generatedDate	Date

6. API Design

Authentication APIs

Method	Endpoint
POST	/auth/register
POST	/auth/login

Vehicle APIs

Method	Endpoint
GET	/vehicles
GET	/vehicles/{id}
POST	/vehicles
PUT	/vehicles/{id}
DELETE	/vehicles/{id}

Booking APIs

Method	Endpoint
POST	/bookings
GET	/bookings
GET	/bookings/{id}
DELETE	/bookings/{id}

Bill APIs

Method	Endpoint
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GET	/bills
-----	--------

GET	/bills/{id}
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7. Frontend Design

Public Pages

- Home Page
 - Login Page
 - Registration Page
-

Customer Pages

- Dashboard
 - Vehicle Listing
 - Vehicle Details
 - My Bookings
 - My Bills
 - Profile
-

Admin Pages

- Dashboard
 - Manage Vehicles
 - Manage Customers
 - Manage Bookings
 - Returns Management
-

8. Security Design

The system uses JWT authentication.

Workflow:

Login



JWT Generated

↓
JWT Stored
↓
JWT Sent with Requests
↓
Backend Validation

Security Features:

- Password Encryption (BCrypt)
 - JWT Authentication
 - Role-Based Access Control
-

9. Future Enhancements

- Online Payment Gateway
 - Email Notifications
 - SMS Notifications
 - Vehicle Maintenance Module
 - Customer Reviews and Ratings
 - Mobile Application
 - GPS Tracking
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10. Conclusion

RentRide is designed as a scalable vehicle rental management platform that automates vehicle booking, availability verification, billing, and customer management. The architecture ensures maintainability, security, and ease of future expansion.